

Who really really owns the Internet?

EXCUSE THE REPETITION OF THE WORD "REALLY" IN THE HEADLINE.

If you Google "who really owns the Internet", you're most likely to find articles pointing to the seven cryptographic keys to the DNSSEC which is managed by ICANN. Yes, those keys do matter but they don't really control the Internet. These keys affect just one function of the Internet. Losing those keys will be a massive inconvenience which is why they're stored in separate guarded facilities that are 4,000 kilometres apart and there are even backup keys which can be used to restore the DNS records should anything happen.

All of this pertains to the DNSSEC and the things that ICANN controls. However, ICANN is only one of the hundreds of stakeholders that are entrusted with the Internet. Think of ICANN as the person standing guard over a bridge which you want to cross. Only that they don't own the bridge and the lord of the estate on which the bridge exists is an entirely different entity altogether. Or should we make that plural? The Internet, as we know it, is a massive network of smaller networks. Then, there are parallel networks which are either isolated entirely from this Internet or are connected at some point either knowingly or anonymously. Some of these connections will lead you to private networks controlled by governments and militaries, some will take you to private networks owned by corporations and some will take you to one of the many shady networks that we collectively term as the dark web. Yes, there's more than one obscure place on the Internet.

The Internet that we're all familiar with is broken down into little fiefdoms which are, in turn, administered by various network operators. Companies that you've most likely never heard of such as Level 3 Parent LLC, Telia Company AB, Cogent Communications and NTT America. Level 3 exercises control over 73.3% of all IP addresses in existence. Telia handles 58.25%, NTT America handles 34.5% and Cogent, 42.79%. There are significant overlaps because of how networks also overlap each other. Since the Internet is a network of networks, there is a mechanism of identifying each network on the Internet. Each network is termed as an Autonomous System (AS) and is assigned an Autonomous

System Number (ASN). As of August 2019, more than 92,000 ASNs have been allocated. Then there's the concept of subletting. Every connected device has an IP address and your home network, which is formed by your router owns all the IP addresses it gives out. However, your router also has an IP address which was lent to it by your ISP which, in turn, got it from another ISP. And that ISP got it from another company and the hierarchy goes on building on top of itself till you get to the big daddies that own the infrastructure. That's where Level 3, Telia and Cogent come into the picture.

Level 3, NTT America and Cogent are American companies whereas Telia is Swedish-Finnish. These are the top four owners of ASNs. Bharti Airtel is the only Indian company in the top 40 largest owners of ASNs and it exercises control over a measly 3.82% of all IP addresses on the Internet. TATA Communications (Rank 43) handles 2.95% and Vodafone (Rank 91) handles 0.32%. Kind of humbling, isn't it? Also, where's Google? The Internet is practically synonymous with this one company.

You might know how all the continents on the Earth are connected via undersea cables? Google owns major shares of the companies that manage these undersea cables. Google partially owns 8.5% and solely owns 1.4% of all undersea cables worldwide. That's 1,02,362 kilometers of submarine cables. The next tech giant in the list is Facebook with 91,850 kilometers of cables to its name, then we have Amazon with 30,557 kms and Microsoft with 6,605. Some of these companies even own satellites and have plans of setting up satellite constellations to control the Internet infrastructure. All of this is just part of the ridiculously convoluted and ever-changing ownership structure of our Internet. It'll be interesting to see how satellite internet constellations such as Starlink, OneWeb and Project Kuiper affect the status quo of ownership.

The Internet is chock full of such facts and I hope this little piece has grabbed your attention because this month's cover story is all about the Internet and is laced with similarly mind-boggling facts. Hopefully, we'll have blown your mind by the end of it. **d**



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